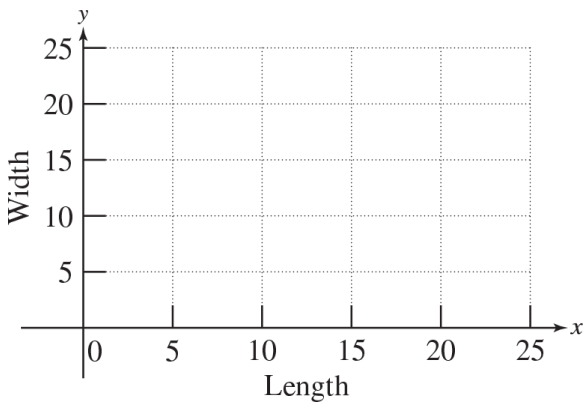
Math 1075
Instructor: Sam Kim
Quiz 8 Name: _____

Section 9.5, 10.1, 10.2
This quiz is worth 10 points and you will have 10 minutes to complete it. Please show all relevant work. **Answers without supporting work will receive no credit**.
Good luck!

Problem 1 (4 points)**.** Suppose Sam has 20 ft of fencing with which he can build a rectangualr garden. Let x represent the length of the garden and let y represent the width.

(a) Write an inequality representing the fact that the total perimeter of the garden is at most 20 ft.

(b) Sketch part of the solution set for this inequality that represents all possible values for the length and width of the garden (Hint: Note that both the length and the width must be positive)



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Problem 2 (3 points)**.** Simplify the expressions. Assume all variables are positive real numbers.

$$\sqrt[4]{81x^{12}y^8}$$

Problem 3 (3 points)**.** Simplify the expressions by using the properties of rational exponents. Write the final answers using positive exponents only.

$$\left(\frac{8x^3y}{27xy^{-6}}\right)^{1/3}$$